NABIL KHAN

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SKILLS

- **Languages:** Python, Java, C/C++, JavaScript, SQL, HTML/CSS, MATLAB
- **Web & Mobile:** React, Node.js, Flask, REST APIs, Android (XML), Git
- **Testing & QA:** TestRail, Jira, JUnit, Test Case Design, Regression Testing, Bug Triage
- **Data & ML:** Scikit-Learn, Pandas, NumPy | **Databases:** PostgreSQL, MongoDB, Firebase
- **Embedded & Hardware:** NI CompactRIO, ESP32, ATmega328P, I2C, PWM, Bluetooth, Verilog, VHDL, Linux

EDUCATION

Concordia University **May 2027**

Bachelor of Engineering, Computer Engineering *Montreal, QC*

- **Coursework:** Data Structures and Algorithms, Operating Systems, Computer Architecture and Design, Digital System Design, Applied Machine Learning, Artificial Intelligence, Fundamentals of Control Systems

WORK EXPERIENCE

Keywords Studios | *QA Intern → QA Analyst* **June 2017 – September 2022**

- Reduced post-release defects by 18% by designing and executing 300+ TestRail test cases across mobile and desktop platforms, expanding regression coverage.
- Accelerated fixes for 500+ critical bugs by writing reproducible Jira reports, triaging severity, and partnering with development teams on targeted solutions.

PROJECTS

Naval Engineering Test Establishment: Breathing Simulation Machine **June 2026 – Present**

- Cut hardware test time by 80%, from 30 to 6 minutes, by building an automated breathing-simulation control system on NI CompactRIO driving linear actuators.
- Delivering live operator visibility into every test run, sampling oxygen and pressure sensors at 100 Hz, by building a real-time frontend and data acquisition pipeline.

Car-cycle: Used Car Price Prediction ML Model **June 2026**

- Extracted 811 encoded valuation features from 161,610 vehicle listings by engineering a Python and Scikit-Learn pipeline to clean, encode, and transform raw data.
- Achieved 0.94 R^2 and $2,068 mean absolute error by benchmarking five regression models with grid search and 5-fold cross-validation, deploying an optimized Random Forest.

Shift Flow: Employee Scheduling Web App **March 2026**

- Generated monthly schedules for 20+ employees in under one minute by building a full-stack JavaScript and MongoDB application with automated shift assignment.
- Cut monthly scheduling errors from 10–15 to 1–2 by building React dashboards that let managers adjust shifts in real time.

Munch: AI Nutrition App & Smart Scale **August 2025 – December 2025**

- Validated real-time Firebase sync with 100+ simulated users by building an Android nutrition-tracking app with a Java backend and XML frontend.
- Cut meal-logging time from 1–2 minutes to 10–15 seconds by engineering an ESP32 smart scale with ±0.01 g precision that streams weights via Bluetooth.

Autonomous Hovercraft: ENGR 290 Team Design Project | 1st Place **Summer 2025**

- Won 1st place among 13 teams in Concordia's autonomous hovercraft maze challenge by co-designing an ATmega328P navigation system fusing IMU, ultrasonic, and infrared sensing.
- Enabled fully autonomous maze navigation by writing embedded C/C++ for IMU-based yaw correction over I2C, ultrasonic wall detection, and PWM fan and servo control.

Brew & Renew: Inventory & POS System **December 2024**

- Streamlined checkout for 20+ consignment products by building a React, Flask, and PostgreSQL POS app with automated QR-code generation and mobile scanning.
- Eliminated spreadsheet settlement errors, from roughly 30 per 100 records to zero, by automating vendor profit-split calculations and inventory updates in Python and SQL.